Head & Neck Anatomy: Temporal & Infratemporal Regions

1. The temporal fossa is a shallow depression on the lateral side of the skull. Its superior boundary is defined by which specific osteological feature?
a. The zygomatic arch
b. The infratemporal crest
c. The suprameatal crest
d. The superior temporal line
e. The coronal suture
2. The floor of the temporal fossa features the pterion, where four bones meet. Deep to this osteological landmark sits which clinically high-yield blood vessel?
a. The deep temporal artery
b. The superficial temporal artery
c. The anterior division of the middle meningeal artery
d. The maxillary artery trunk
e. The internal carotid artery
3. The temporal fossa is securely filled by the temporalis muscle. This muscle is functionally classified as a muscle of mastication and arises from the bony floor and the:
a. Temporal fascia
b. Zygomatic arch inner table
c. Pterygoid hamulus
d. Lateral pterygoid plate
e. Pterygomandibular raphe
4. The infratemporal fossa is an irregularly shaped space situated deep and inferior to the zygomatic arch. The roof of this fossa is formed primarily by the infratemporal surface of the greater wing of the sphenoid bone and features which major opening?
a. Foramen rotundum
b. Foramen ovale
c. Foramen lacerum
d. Jugular foramen
e. Optic canal
5. The anterior wall of the infratemporal fossa is formed by the posterior, non-articular surface of which bone?
a. The zygomatic bone
b. The maxilla (maxillary tuberosity)
c. The palatine bone vertical plate
d. The lateral pterygoid plate
e. The ramus of the mandible
6. The medial wall of the infratemporal fossa is a vital landmark that provides anchor points for masticatory muscles. This wall is formed by the:
a. Medial pterygoid plate
b. Lateral surface of the lateral pterygoid plate
c. Styloid process of the temporal bone
d. Vertical plate of the palatine bone
e. Posterior surface of the maxilla

7. The lateral wall of the infratemporal fossa forms its outer boundary and is accessible during extraoral clinical procedures. This wall is composed of the:
a. Zygomatic bone facial surface
b. Condylar process of the mandible
c. Coronoid process and the medial surface of the ramus of the mandible
d. Squamous part of the temporal bone
e. Infratemporal crest of the sphenoid
8. The infratemporal fossa communicates directly with the pterygopalatine fossa medially through a vertical, fissure-like opening called the:
a. Superior orbital fissure
b. Inferior orbital fissure
c. Sphenopalatine foramen
d. Pterygomaxillary fissure
e. Foramen rotundum
9. The infratemporal fossa communicates superiorly with the orbit cavity through which specific horizontal bony opening?
a. Superior orbital fissure
b. Pterygomaxillary fissure
c. Optic canal
d. Foramen rotundum
e. Inferior orbital fissure
10. Muscles of mastication drive mandibular movements. The temporalis muscle inserts into the margins and medial surface of which process of the mandible?
a. Condylar process
b. Styloid process
c. Alveolar process
d. Mental tubercle
e. Coronoid process
11. The powerful masseter muscle originates from the lower border and medial surface of the zygomatic arch. It inserts into the lateral surface of the:
a. Coronoid process
b. Condylar neck
c. Pterygoid fovea
d. Ramus and angle of the mandible
e. Mental protuberance
12. The lateral pterygoid muscle possesses two distinct heads of origin. The superior head originates from the infratemporal surface of the greater wing of the sphenoid, while the inferior head arises from the:
a. Medial surface of the medial pterygoid plate
b. Lateral surface of the lateral pterygoid plate
c. Maxillary tuberosity
d. Pyramidal process of the palatine bone
e. Pterygoid hamulus
13. The lateral pterygoid muscle inserts into the pterygoid fovea on the anterior neck of the condyle and directly into the:
a. Sphenomandibular ligament
b. Capsule and articular disc of the temporomandibular joint (TMJ)
c. Stylomandibular ligament

d. Retromolar triangle meshwork
e. Mylohyoid ridge

14. Which muscle of mastication is functionally unique because it acts as the primary opener of the jaw by pulling the mandibular condyle forward?

a. Masseter muscle
b. Medial pterygoid muscle
c. Temporalis muscle
d. Lateral pterygoid muscle
e. Anterior belly of digastric

15. The medial pterygoid muscle also possesses two heads. Its large deep head originates from the medial surface of the lateral pterygoid plate, and its small superficial head arises from the:

a. Infratemporal crest
b. Maxillary tuberosity and pyramidal process of the palatine bone
c. Body of the sphenoid bone
d. Pterygoid hamulus apex
e. Mandibular condyle neck

16. The medial pterygoid muscle runs downward, backward, and laterally to insert into the roughened medial surface of the mandible near the:

a. Coronoid process tip
b. Condylar neck fovea
c. Alveolar border
d. Mental foramen entrance
e. Angle of the mandible

17. All four principal muscles of mastication develop from the same embryonic source and are innervated by motor fibers traveling within the:

a. Maxillary nerve (V2)
b. Ophthalmic nerve (V1)
c. Mandibular nerve (V3 division of Trigeminal)
d. Facial nerve (CN VII)
e. Glossopharyngeal nerve (CN IX)

18. The mandibular nerve (V3) enters the infratemporal fossa from the middle cranial fossa by passing vertically down through which opening?

a. Foramen rotundum
b. Foramen ovale
c. Foramen spinosum
d. Foramen lacerum
e. Jugular foramen

19. Immediately below the foramen ovale, the main trunk of the mandibular nerve gives off a small recurrent branch that re-enters the cranium via the foramen spinosum. This is the:

a. Nervus spinosus (meningeal branch)
b. Deep temporal nerve
c. Auriculotemporal nerve
d. Buccal nerve
e. Chorda tympani nerve

20. The main trunk of V3 also gives off a motor branch to which muscle that tenses the soft palate during swallowing?

a. Tensor tympani muscle
b. Tensor veli palatini muscle
c. Medial pterygoid muscle
d. Levator veli palatini muscle
e. Stylopharyngeus muscle

21. The anterior division of the mandibular nerve is primarily motor. However, it gives off one essential sensory branch that emerges between the two heads of the lateral pterygoid muscle to supply the skin and mucous membrane of the cheek. This is the:

a. Lingual nerve
b. Buccal nerve (long buccal nerve)
c. Inferior alveolar nerve
d. Mental nerve
e. Auriculotemporal nerve

22. The long buccal nerve provides sensory supply to the cheek mucosa. It must not be confused with the buccal branch of the facial nerve (CN VII), which provides:

a. Sensory supply to the lower lip
b. Motor supply to the buccinator and facial muscles
c. Parasympathetic supply to the parotid gland
d. Taste sensation to the anterior tongue
e. Motor supply to the masseter muscle

23. The posterior division of the mandibular nerve is primarily sensory. It splits into three major branches, which are the:

a. Deep temporal, masseteric, and lateral pterygoid nerves
b. Auriculotemporal, lingual, and inferior alveolar nerves
c. Infraorbital, zygomatic, and superior alveolar nerves
d. Greater palatine, lesser palatine, and nasopalatine nerves
e. Mental, incisive, and mylohyoid nerves

24. The auriculotemporal nerve originates by two distinct roots that encircle a major middle cranial artery before consolidating. This artery is the:

a. Maxillary artery trunk
b. Superficial temporal artery
c. Middle meningeal artery
d. Deep temporal artery
e. Internal carotid artery

25. The auriculotemporal nerve ascends behind the TMJ to supply the skin of the temple and ear. It also transmits postganglionic parasympathetic secretomotor fibers to the:

a. Submandibular salivary gland
b. Sublingual salivary gland
c. Parotid salivary gland
d. Lacrimal gland
e. Labial minor glands

26. The parasympathetic secretomotor fibers heading to the parotid gland synapse in which peripheral ganglion located in the infratemporal fossa, directly medial to V3?

a. Ciliary ganglion
b. Pterygopalatine ganglion
c. Submandibular ganglion
d. Otic ganglion

e. Geniculate ganglion

27. The lingual nerve runs downward and forward in the infratemporal fossa. It is joined at an acute angle from behind by a branch of the facial nerve carrying taste and parasympathetic fibers, which is the:

a. Greater petrosal nerve
b. Lesser petrosal nerve
c. Chorda tympani nerve
d. Deep petrosal nerve
e. Tympanic plexus nerve

28. The chorda tympani nerve carries special visceral afferent (taste) fibers from the:

a. Posterior one-third of the tongue
b. Anterior two-thirds of the tongue
c. Hard palate mucosa
d. Epiglottis region
e. Circumvallate papillae row

29. The inferior alveolar nerve enters the mandibular foramen to run inside the mandibular canal. Before entering the bone, it gives off which pure motor branch?

a. Lingual nerve branch
b. Mental nerve branch
c. Mylohyoid nerve (supplying mylohyoid and anterior belly of digastric)
d. Masseteric nerve branch
e. Deep temporal branch

30. The maxillary artery is one of the terminal branches of the external carotid artery. It originates within the parenchyma of the parotid gland at the neck of the:

a. Styloid process
b. Mastoid process
c. Mandible condyle
d. Infratemporal crest
e. Lateral pterygoid plate

31. The first (mandibular) part of the maxillary artery runs horizontally behind and deep to the neck of the condyle. A major branch that ascends through the foramen spinosum to supply the dura mater is the:

a. Deep temporal artery
b. Inferior alveolar artery
c. Accessory meningeal artery
d. Middle meningeal artery
e. Masseteric artery

32. The second (pterygoid) part of the maxillary artery runs obliquely forward and upward, either superficial or deep to which muscle of mastication?

a. Medial pterygoid muscle
b. Lateral pterygoid muscle
c. Masseter muscle
d. Temporalis muscle
e. Buccinator muscle

33. The second part of the maxillary artery is characterized by giving off branches that are purely:

a. Meningeal in distribution
b. Intracranial in destination

c. Muscular branches supplying the muscles of mastication and buccinator
d. Glandular branches to the parotid
e. Cutaneous branches to the forehead

34. The third (pterygopalatine) part of the maxillary artery leaves the infratemporal fossa by passing through the pterygomaxillary fissure to enter the:
a. Orbit cavity
b. Nasal cavity proper
c. Pterygopalatine fossa
d. Temporal fossa
e. Middle cranial fossa

35. Enmeshed around and within the lateral pterygoid muscle is a massive vascular network of valveless veins known as the:
a. Cavernous venous network
b. Pterygoid venous plexus
c. Pharyngeal venous plexus
d. Submandibular plexus
e. Internal vertebral plexus

36. The pterygoid venous plexus communicates directly with the intracranial cavernous sinus through emissary veins passing through the:
a. Foramen ovale and foramen lacerum
b. Foramen rotundum and optic canal
c. Foramen spinosum exclusively
d. Jugular foramen
e. Hypoglossal canal

37. The pterygoid venous plexus drains posteriorly into the maxillary vein, which joins the superficial temporal vein to form the:
a. Internal jugular vein
b. External jugular vein
c. Retromandibular vein
d. Anterior jugular vein
e. Common facial vein

38. The sphenomandibular ligament is an accessory ligament of the TMJ that represents a remnant of Meckel's cartilage. Superiorly, it attaches to the spine of the sphenoid; inferiorly, it inserts into the:
a. Condylar neck fovea
b. Angle of the mandible
c. Lingula of the mandible
d. Mylohyoid line
e. Coronoid process edge

39. The temporomandibular joint (TMJ) is a synovial joint of the condylar variety. Its internal cavity is completely divided into superior and inferior compartments by an:
a. Fibrous capsule sleeve
b. Articular disc (composed of fibrocartilage)
c. Lateral ligament band
d. Extracapsular synovial fold
e. Sphenomandibular band

40. Gliding movements (translation) of the TMJ, where the condyle moves forward onto the articular eminence during wide jaw opening, take place exclusively within the:
a. Lower joint compartment
b. Upper joint compartment
c. Pterygoid fovea zone
d. Mandibular notch axis
e. Infratemporal roof matrix
41. Pure rotational movements (hinge movements) of the TMJ during the initial phases of jaw opening take place entirely within the:
a. Upper joint compartment
b. Lower joint compartment
c. Retrodiscal tissue pads
d. Lateral pterygoid tendon fibers
e. Infratemporal crest plane
42. To perform an inferior alveolar nerve block for dental extractions, the needle is passed through the oral mucosa lateral to the pterygomandibular raphe. The solution must be deposited near the:
a. Mental foramen
b. Mandibular foramen (near the lingula)
c. Infraorbital foramen
d. Pterygoid fovea
e. Infratemporal crest roof
43. If the dental needle is accidentally directed too far posteriorly during an inferior alveolar nerve block, the anesthetic solution will enter the parotid gland capsule and block the facial nerve (CN VII), causing:
a. Total numbness of the tongue
b. Severe pain in the temporal fossa
c. Transient facial paralysis (inability to close the eye or smile on that side)
d. Profuse salivation from the parotid duct
e. Spasm of the masseter muscle (trismus)
44. A patient experiences a severe spasm of the muscles of mastication, making it impossible to open the mouth. This clinical sign is known as:
a. Epiphora
b. Trismus (lockjaw)
c. Miosis
d. Ptosis
e. Ophthalmoplegia
45. Infection can spread from a diseased lower molar tooth retrogradely into the pterygoid venous plexus and track into the cavernous sinus, causing cavernous sinus thrombosis. This tracking occurs because these veins:
a. Contain high-pressure muscular pumps
b. Are held permanently open by bone tissue
c. Are valveless, allowing backward flow of septic emboli
d. Connect directly to the lateral ventricles
e. Open into the subarachnoid space
46. The masseteric nerve passes from the infratemporal fossa to the masseter muscle by traveling through which osteological pathway?

a. Foramen ovale
b. Foramen spinosum
c. Mandibular notch
d. Pterygomaxillary fissure
e. Inferior orbital fissure

47. The deep temporal nerves, which supply the powerful temporalis muscle, are direct branches arising from the:
a. Main trunk of V3
b. Anterior division of the mandibular nerve (V3)
c. Posterior division of the mandibular nerve (V3)
d. Maxillary nerve (V2)
e. Facial nerve temporal branch

48. The otic ganglion sits structurally directly below the foramen ovale. The preganglionic parasympathetic fibers that terminate in this ganglion are carried by the lesser petrosal nerve from which cranial nerve?
a. Oculomotor nerve (CN III)
b. Facial nerve (CN VII)
c. Glossopharyngeal nerve (CN IX)
d. Vagus nerve (CN X)
e. Trigeminal nerve (CN V)

49. Postganglionic sympathetic fibers pass through the otic ganglion without synapsing to supply the blood vessels of the parotid gland. These sympathetic fibers originate in the:
a. Ciliary ganglion
b. Pterygopalatine ganglion
c. Superior cervical ganglion
d. Middle cervical ganglion
e. Stellate ganglion

50. During extraoral surgical relocation of a fractured mandibular condyle, a clinician must be extremely careful to protect the vessel that climbs vertically just behind the neck of the condyle, which is the:
a. Maxillary artery trunk
b. Superficial temporal artery
c. Facial artery
d. Middle meningeal artery
e. Lingual artery

Answer Key Table (Temporal & Infratemporal Fossae - Sequence 2)

Q 1-8	Q 9-16	Q 17-24	Q 25-32	Q 33-40	Q 41-48	Q 49-50
d	**e**	**c**	**c**	**c**	**b**	**c**

c	**e**	**b**	**d**	**c**	**b**	**b**
a	**b**	**a**	**c**	**b**	**c**	
b	**b**	**b**	**b**	**c**	**b**	
b	**b**	**d**	**b**	**a**	**c**	
b	**e**	**c**	**a**	**b**	**c**	
c	**c**	**d**	**a**	**c**	**b**	
d	**e**	**d**	**d**	**b**	**c**	